

SHEBOYGAN, WI, USA

WMO: 994330

Lat: 43.7495N

Lon: 87.6927W

Elev: 577

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	2.2	-13.3	2.7	-1.7	-8.2	3.6	3.4	47.2	30.6	39.5	29.8	16.6	280	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	83.0	71.1	79.5	70.3	76.6	69.6	76.1	78.9	73.8	76.7	71.8	75.0	15.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	135.2	77.1	73.0	125.2	75.4	70.9	116.3	73.5	39.9	79.2	37.8	76.7	36.0	75.1	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.5	34.1	28.7	-8.3	89.8	6.6	3.9	-13.0	92.6	-16.9	94.8	-20.6	97.0	-25.4	99.8
			-7.8	78.5	5.7	3.2	-11.9	80.8	-15.2	82.6	-18.4	84.4	-22.5	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	21.9	24.4	33.1	42.6	51.1	61.3	68.4	68.4	61.9	50.7	39.4	28.0
	DBStd	18.01	11.34	9.90	8.74	6.44	6.78	6.80	5.67	5.01	6.40	7.05	8.90	10.17
	HDD50	3515	872	716	526	238	68	2	0	0	3	78	329	683
	HDD65	7259	1337	1136	988	673	433	150	29	22	131	444	768	1148
	CDD50	2074	0	0	3	14	103	341	570	569	361	101	11	1
	CDD65	343	0	0	0	0	3	39	134	126	39	2	0	0
	CDH74	1275	0	0	0	4	38	162	588	387	89	7	0	0
	CDH80	270	0	0	0	0	7	26	146	75	16	0	0	0
Wind	WSAvg	14.0	15.5	16.1	15.5	15.7	14.1	11.3	11.2	11.4	12.6	14.3	14.9	15.1
Precipitation	PrecAvg	31.6	1.9	1.5	1.8	3.2	3.7	4.2	3.4	3.7	2.7	2.4	2.3	1.6
	PrecMax	40.6	3.0	4.9	4.5	5.6	11.1	10.7	6.5	10.1	5.9	5.2	5.9	3.3
	PrecMin	25.9	0.4	0.3	0.3	1.1	1.0	0.9	1.4	1.4	0.2	0.8	0.4	0.3
	PrecStd	4.1	0.7	1.1	1.0	1.2	2.2	2.7	1.3	2.1	1.4	1.4	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	46.2	62.3	72.4	79.8	83.5	89.7	87.3	81.7	73.7	63.0	52.5
		MCWB	42.6	41.6	53.4	58.4	63.3	68.7	76.3	74.5	70.6	61.4	55.0	49.4
	2%	DB	42.2	41.4	52.4	60.5	70.5	78.7	83.9	81.7	76.1	66.6	57.2	46.5
		MCWB	39.1	38.6	45.1	51.4	60.2	69.1	71.5	71.8	68.3	61.0	52.5	43.8
	5%	DB	39.0	38.8	47.5	54.4	65.6	75.1	80.1	77.8	73.3	63.3	53.8	43.2
		MCWB	37.0	36.9	42.6	47.7	57.9	66.6	70.5	71.5	67.9	58.8	49.9	41.0
	10%	DB	36.4	36.7	43.3	50.1	61.2	71.4	76.8	75.4	71.0	60.5	50.7	40.7
		MCWB	35.0	35.1	39.9	44.7	55.2	65.3	69.7	70.9	66.5	56.6	47.6	38.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	43.2	55.9	59.4	68.2	74.6	79.3	78.9	74.3	66.1	56.7	50.6
		MCDB	45.0	45.7	58.7	67.6	71.9	77.8	81.4	82.4	76.1	69.1	61.0	52.1
	2%	WB	39.5	39.4	45.9	52.8	62.9	71.8	75.7	76.5	71.4	62.9	53.9	45.5
		MCDB	41.1	41.0	49.7	58.0	68.9	75.4	80.1	78.8	74.4	65.2	56.8	46.4
	5%	WB	37.1	37.3	42.1	48.4	59.0	69.0	73.3	74.6	69.7	60.1	51.1	41.6
		MCDB	38.5	38.6	44.7	52.0	63.4	71.9	77.1	76.3	72.5	62.1	53.2	42.6
	10%	WB	34.9	35.3	39.4	45.1	55.5	66.5	71.3	72.4	67.8	57.7	48.3	38.8
		MCDB	36.3	36.7	42.0	48.7	59.7	69.9	75.1	74.7	70.3	59.9	50.2	40.3
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.6	10.2	10.0	11.4	11.5	10.9	9.4	10.3	10.1	9.9	10.3
		MCDBR	12.3	10.4	16.7	19.5	20.0	17.8	16.3	13.1	13.4	14.0	13.6	11.6
		MCWBR	11.1	9.7	12.1	13.1	13.4	11.5	9.7	8.7	9.3	10.2	11.0	10.2
	5% WB	MCDBR	10.5	10.3	14.4	18.2	18.7	15.7	14.5	11.0	11.5	11.9	12.3	10.5
		MCWBR	10.1	9.8	11.9	13.5	14.8	12.5	10.9	8.6	9.3	10.7	11.1	10.0
Clear-Sky Solar Irradiance	taub	0.286	0.314	0.338	0.385	0.409	0.427	0.419	0.413	0.388	0.345	0.309	0.286	
	taud	2.447	2.349	2.385	2.297	2.286	2.286	2.312	2.322	2.376	2.488	2.522	2.490	
	Ebn at Noon	270	278	285	279	274	269	269	267	265	265	259	259	
	Edh at Noon	23	30	33	39	41	41	40	38	34	26	22	20	
All-Sky Solar Radiation	RadAvg	462	762	1138	1439	1706	1899	1949	1666	1331	842	521	366	
	RadStd	62	84	109	135	142	132	86	107	107	92	46	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page